

Murray S. Bennett

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Research Summary and Interests

My research focuses on decision-making under uncertainty in applied settings, particularly where human judgement interacts with intelligent systems. Using formal cognitive models and computational simulations, I study perceptual integration, workload, and metacognition in tasks such as medical image interpretation and collaborative decision-making. A central goal of my work is to identify principled ways to improve performance in human-AI teams.

EDUCATION

Ph.D. Cognitive Psychology 2023
University of Newcastle, Australia
Collective Workload: Human Performance in Competitive and Collaborative Tasks

B.S. Psychology (*1st Class Honours*) 2018
University of Newcastle, Australia
Honours Thesis: The Mental Representation of Symbolic Numerals

RESEARCH

Postdoctoral Scholar 2024-
The Ohio State University, Department of Psychology

- Develop computational methods for communicating neural network metacognition to human users, with the goal of improving human-AI collaboration and decision-making.
- Apply advanced quantitative cognitive models (including Systems Factorial Technology and General Recognition Theory) to test processing independence and interactions in social and cognitive decision-making tasks.

Postdoctoral Researcher 2022-24
UT San Antonio, Psychology Department

- Quantified and combined human perceptual judgments with computer-vision outputs to demonstrate performance gains in hybrid human-AI melanoma classification.
- Used General Recognition Theory to examine perceptual interactions and feature integration in medical image decision-making.

Data Analyst*2021-22*

Hunter New England Health Local Health District

- Conducted weekly statistical analyses of nationwide FluTracking and COVID-19 survey data for Australian federal government reporting.
- Collaborated with international partners to design, implement, and maintain large-scale survey infrastructure for data collection in New Zealand and Argentina.

Research Assistant*2021*

University of Newcastle, School of Biomedical Sciences and Pharmacy

- Designed, conducted, and analysed experiments examining physiological correlates of cognitive workload using the Detection Response Task.

Research Assistant*2019-22*

University of Newcastle, School of Psychology

- Provided technical and methodological support for research on cognitive processing, decision-making, and perceptual judgement.
- Designed and implemented online versions of foundational cognitive psychology experiments for research and undergraduate instruction.

Research Internship*2018*

University of Newcastle, Centre for Brain and Mental Health Research

- Analysed cognitive workload and flight-simulator performance data from helicopter pilots evaluating novel 3D helmet display systems.
- Examined relationships between task demands, performance, and subjective workload in applied human-factors contexts.

TEACHING

Guest Lecturer

Artificial Intelligence in Psychology

OSU 2026

- Lecture and supervising student-lead presentations on k-means clustering.

Course Instructor

Introduction to Psychology

UT San Antonio 2024

- Introduction to the study of and scientific approaches to attention, sensation, perception, emotion, learning, memory, problem solving, personality, development, atypical behaviour, and social behaviour.

Honours Thesis Grader

University of Newcastle, School of Psychology

UoN 2021

- Assessed and critiqued the methodology, data analysis, and scientific writing of honours-level cognitive psychology research projects.

Lab Demonstrator

Advanced Research Methods & Design

UoN 2021

- Presenting practical content and guiding students in the hands-on application of advanced concepts in research methodology, statistics, Bayesian analysis, and experimental design.

Cognitive Psychology

UoN 2020

- Introduction and overview of fundamental cognitive psychology principles, including attention, memory, language, reasoning, and problem solving, with focus on the application of experimental methods and formal psychology report writing.

Accreditation

Associate *Centre for the Integration of Research, Teaching and Learning*, 2024

- Associate-level teaching accreditation via CIRTl by completing professional development events and demonstrating expertise in evidence-based teaching.

PROFESSIONAL SERVICE

Conference Session Chair

2022, 2025, 2026

Society for Mathematical Psychology (2025, 2026)

Australian Society for Mathematical Psychology (2022)

Conference Reviewing and Judging

2023–2026

Graduate abstracts and presentations judge, Hayes Advanced Research Forum (OSU)

Abstract reviewer, Southwestern Psychological Association

3-Minute Thesis judge, UT San Antonio

UNIVERSITY SERVICE

Founder and Chair, Postdoctoral Writing Club

2024

UT San Antonio

Postdoctoral Service Roles

2023–2024

Postdoctoral Appreciation Week Organizing Committee, UT San Antonio

Postdoctoral Mentor of the Year Judge, UT San Antonio

Postdoctoral Association Member, UT San Antonio

Community and Governance Service

2020–2021

Board of Directors, Community Representative, Newcastle University Sport

PUBLICATIONS

Peer-reviewed journal articles

- Hedley, L. G., Russel, L., **Bennett, M. S.**, and Eidels, A. (In Press). Benchmarking the benchmark: A novel simulation benchmark for team performance. *Chinese Journal of Psychology*
- Kerr, A., Carlson, S., Baldwin, Z., Innes, R., Howard, Z., **Bennett, M. S.**, Evans, T., McKenzie, T., and Dalton, C. (2025). Flutracking: Weekly online community-based surveillance of respiratory illness in australia, 2020–2022 report. *Communicable Diseases Intelligence*, 49
- Gronau, Q. F., **Bennett, M. S.**, Brown, S. D., Hawkins, G. E., and Eidels, A. (2023). Do choice tasks and rating scales elicit the same judgments? *Journal of Choice Modelling*, 49:100437
- Bennett, M. S.** (2023). Collective workload: Human performance in competitive and collaborative tasks. *NOVA repository*
- Bennett, M. S.**, Hedley, L. G., Love, J., Houpt, J. W., Brown, S. D., and Eidels, A. (2022). Human performance in competitive and collaborative human–machine teams. *Topics in Cognitive Science*
- John, A. R., Singh, A. K., Do, T.-T. N., Eidels, A., Nalivaiko, E., Gavgani, A. M., Brown, S. D., **Bennett, M. S.**, Lal, S., Simpson, A. M., et al. (2022). Unraveling the physiological correlates of mental workload variations in tracking and collision prediction tasks. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 30:770–781
- Garrett, P. M., **Bennett, M. S.**, Hsieh, Y.-T., Howard, Z. L., Yang, C.-T., R., D., and Eidels, A. (2022). Wheel of fortune: A cross-cultural examination of how expertise shapes the mental representations of familiar and unfamiliar numerals. *Computational Brain & Behavior*, 5(1):45–59
- Bennett, M. S.**, Mullard, R., Adam, M. T., Steyvers, M., Brown, S. D., and Eidels, A. (2020). Going, going, gone: competitive decision-making in dutch auctions. *Cognitive Research: Principles and Implications*, 5(1):1–22

Under Review

- Bennett, M. S.** and Kvam, P. (Under review). Machine metacognition improves classification performance and uncertainty quantification. *Preprint*, <https://osf.io/f34rm>
- Bennett, M. S.**, Alaukik, A., Smith, C. T., and Kvam, P. D. (Under reviewa). Assessing process independence in implicit attitudes: An sft and grt modeling analysis.
- Bennett, M. S.**, Houpt, J. W., Wenger, M. J., Townsend, J. T., and Mark, L. (Under reviewb). Enhancing melanoma detection by combining continuous feature estimates from human perception and computer vision. *Preprint*, <https://osf.io/ctg9s>

Conference Proceedings

- Hedley, L. G., **Bennett, M. S.**, Love, J., Houpt, J. W., Brown, S. D., and Eidels, A. (2023). The relationship between teaming behaviours and joint capacity of hybrid human-machine teams. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, volume 45

Reviewing Activity

Ad hoc reviewer for: *Cognitive Research: Principles and Implications, Computational Brain and Behaviour, Topics in Cognitive Science, Communications Psychology, Journal of Mathematical Psychology*

PRESENTATIONS

Invited

Human-AI perceptual judgments of malignant melanoma. 2024, Psychology Department, The University of Newcastle, Australia.

Human-AI collaboration. 2024, Department of Psychological Sciences at Tarleton State University.

Collaborative and competitive 'Team Spirit'. 2024, Joint meeting of the Newcastle Cognition Lab and UC Irvine's MADLAB.

Talks

Bennett, M. S., Alaukik, A., Smith, C., and Kvam, P. (2025, July 25-28). Processing independence in implicit attitudes: An sft and grt analysis. In *Meeting of the Society for Mathematical Psychology/ICCM*

Bennett, M. S. and Houpt, J. W. (2024, June 17-21). A grt examination of the abcs for melanoma identification with comparison to a deep convolutional neural network. In *Virtual Meeting of the Society for Mathematical Psychology/ICCM*

Bennett, M. S. and Houpt, J. W. (2024, March 27-30). Using ai and human feature perception to improve melanoma identification. In *Southwestern Psychology Association Convention*

Bennett, M. S. and Houpt, J. W. (2023, October 6-7). Improving human perception and computer vision identification of malignant melanoma. In *The Association for Research in Memory, Attention, Decision-making, Imaging, Language, Learning, and Organization*

Bennett, M. S. and Houpt, J. W. (2023, July 18-21). A bradley-terry-luce model-based analysis of human perception and computer vision to identify melanoma lesions. In *Meeting of the Society for Mathematical Psychology*

Bennett, M. S. (2021, April 9-11). Cognitive workload and performance of collaborative and competitive teams. In *Australasian Experimental Psychology Society*

Bennett, M. S. (2021, February 11-12). Cognitive workload and performance of collaborative and competitive teams. In *Australasian Mathematical Psychology Conference, Newcastle*

Bennett, M. S. (2020, October 16). Effects of competitive and collaborative team environments on individual cognitive load. In *Centre for Brain and Mental Health Research Post-Graduate and Postdoctoral Conference, Newcastle*

Bennett, M. S. (2020, July 20-31). An iterated prospect theory model for the dutch auction. In *Society for Mathematical Psychology/ICCM Virtual Conference*

Bennett, M. S., Mullard, R., Adam, M. T., Steyvers, M., Brown, S. D., and Eidels, A. (2020, February 12-14). Competitive decision-making in dutch auctions. In *Australasian Mathematical Psychology Conference, Sydney*

Eidels, A. and **Bennett, M. S.** (2019, November 13). Workload capacity in collaborative and competitive group performance. In *Configural Processing Consortium, Montréal*

Bennett, M. S. and Eidels, A. (2019, August 2). Mental workload in a team based dual task. In *Centre for Brain and Mental Health Research Post-Graduate Conference, Newcastle*

Bennett, M. S., Eidels, A., and Garrett, P. M. (2019, February 14-16). The cost of errors: Confusion analysis and the mental representation of numerals. In *Australasian Mathematical Psychology Conference, Melbourne*

Bennett, M. S., Garrett, P. M., Yang, C.-T., and Eidels, A. (2018, November 14). Mental representation of familiar and unfamiliar numerals. In *Configural Processing Consortium, New Orleans*

Bennett, M. S. and Eidels, A. (2018, November 9). At sixes and sevens: The mental representation of familiar and unfamiliar symbolic numerals. In *University of Newcastle Psychology Student Research Conference, Ourimbah*

Posters

Bennett, M., S. (2026, March 03). Latent space density as a proxy for confidence in neural networks. OSU AI Research Summit

Bennett M., S. (2025, November 20). Latent space density as a proxy for confidence in neural networks. MathPsych at Psychonomics Satellite Symposium

Bennett M., S. (2025, November 20-23). Integrating human and ai perception for enhanced melanoma diagnosis. *66th Annual Meeting of the Psychonomic Society*

Bennett M., S. (2023, September 19). Improving human perception and computer vision identification of malignant melanoma. *San Antonio Postdoctoral Research Forum*

Bennett, M. S. (2023, September 22). Human perception of melanoma features vs computer vision algorithm. *University of Texas at San Antonio Postdoctoral Showcase*

PROFESSIONAL SOCIETIES

Society for Mathematical Psychology	2019-present
Psychonomic Society	2019-present
Southwestern Psychological Association	2024